

A project report on

**AI DRIVEN - DRIVER'S DROWSINESS
DETECTION SYSTEM**

Submitted in partial fulfilment for the award of the degree of

Bachelor of Technology in

COMPUTER SCIENCE AND ENGINEERING

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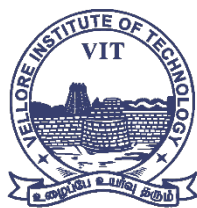
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ABSTRACT

Road safety remains a critical global challenge, with driver fatigue and drowsiness accounting for a significant percentage of fatal accidents. Traditional safety mechanisms often fail to detect the subtle, early physiological signs of exhaustion, such as micro-sleeps or reduced situational awareness. This project proposes an **Intelligent Driver's Drowsiness Detection System** designed to provide real-time monitoring and active intervention.

Leveraging computer vision and deep learning, the system utilizes a camera to continuously track facial landmarks. The core detection logic is built on two primary metrics: the **Eye Aspect Ratio (EAR)** to identify prolonged eye closure and the **Mouth Aspect Ratio (MAR)** to detect frequent yawning. To provide a more holistic safety profile, the system also incorporates **emotion detection** to monitor psychological stressors like anger or frustration.

Developed using **Python, OpenCV, and Dlib**, the software is integrated with **Arduino** hardware to trigger immediate physical alerts, including buzzers and LED indicators. In the event of driver unresponsiveness, the system is capable of initiating an emergency **SOS protocol** with location-based messaging. The results demonstrate a high-accuracy, low-latency solution that significantly enhances road safety by transitioning from passive observation to active, life-saving prevention.

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LIST OF ABBREVIATIONS

- **AI** – Artificial Intelligence
- **EAR** – Eye Aspect Ratio
- **MAR** – Mouth Aspect Ratio
- **SOS** – Save Our Souls (Emergency alert system)
- **GPS** – Global Positioning System
- **API** – Application Programming Interface
- **CPU** – Central Processing Unit
- **IR** – Infrared
- **IoT** – Internet of Things
- **ECU** – Electronic Control Unit
- **EEG** – Electroencephalography
- **EOG** – Electrooculography
- **ECG** – Electrocardiography
- **FACS** – Facial Action Coding System
- **PERCLOS** – Percentage of Eye Closure
- **YOLO** – You Only Look Once (object detection model)
- **HOG** – Histogram of Oriented Gradients
- **LCD** – Liquid Crystal Display
- **USB** – Universal Serial Bus
- **fps** – Frames per second
- **ms** – Milliseconds

Chapter 1

Introduction

1.1 Overview of Global Road Safety

One of the most important issues of the contemporary society is road safety. Traffic accidents have become the prime cause of injury and death with the world witnessing a high rate of growth in the number of vehicles on the road. The drowsiness and fatigue of drivers are among the most hazardous and underestimated reasons of accidents on the road. According to one of the studies conducted by the AAA Foundation for Traffic Safety, about 17.6% of all the fatality crashes in the year 2017 to 2021 involved a drowsy driver, which led to the number of 29,834 deaths in the same period (Tefft, 2024).

Drowsiness decreases the reaction time, judgement, and a driver cannot make fast decision. In comparison to other factors that include speeding and drunk driving, drowsiness is sometimes hard to notice on the outside and thus it is a silent killer. Automated systems can be of great use because drivers can be unaware of when they are leaving an alert state and entering the state of fatigue.

The conventional methods of driver fatigue monitoring are steering pattern recognition, lane distraction, and wearable (Lee et al., 2016; Jia et al., 2022). Nevertheless, these are either inaccurate, or manufacture intrusive hardware (hardware that needs some kind of physical interaction). Due to the development of Artificial Intelligence (AI) and Computer Vision, now, however, it is possible to create non-invasive, real-time driver observing machine systems based on cameras and machine learning models.

Our solution to this situation is to create a drowsiness detector, which would work in real time, equipped to monitor facial features and eye movement of the driver along with his/her emotion using a basic webcam. This system works on a visual data processing model to identify alert or fatigued driver based on visual information and computer vision algorithms and machine learning.

As opposed to a traditional system, the following features are incorporated in this project:

- Eye Aspect Ratio (EAR) for blink detection

- Mouth Aspect Ratio (MAR) for yawning detection
- Emotional state analysis using deep learning
- Hardware integration for alerts
- Emergency SOS notification system

The main goal of the project is to minimize the accidents due to fatigue of drivers by offering prompt warnings and interventions. The system will be cheap, easily scalable, and implementable by using available hardware and open-source technologies, which can be used in real-life applications.

1.2 SYSTEM OVERVIEW

Our AI enables us to keep track of the driver behaviour through the integration of computer vision, machine learning, and hardware. The system is live and receives video data of a webcam to identify the symptoms of fatigue and drowsiness.

1.2.1 OVERALL ARCHITECTURE

The system can be divided into the following major components:

1. Face and Landmark Detection
2. Feature Extraction (EAR & MAR)
3. Emotional Processing Pipeline
4. Threshold Logic and Decision System
5. Alert and Hardware Integration
6. Live Dashboard and Logging System
7. Emergency SOS Module

All the elements collaborate with each other to provide proper detection and early response.

1.2.2 FACE AND LANDMARK DETECTION

The initial process in the pipeline is to identify the face of the driver using a live video feed. It is done with the means of OpenCV and face landmark libraries like Dlib.

Depending on each frame, the system finds the face part and isolates 68 facial locations, comprising of key points that surround:

- Eyes
- Mouth
- Nose
- Jawline

These landmarks are crucial for calculating eye closure and mouth opening, which are indicators of fatigue.

1.3 MATHEMATICAL RATIO EXTRACTION (EAR & MAR)

After identification of facial landmarks, the system computes the following ratios:

1.3.1 EYE ASPECT RATIO (EAR)

The Eye Aspect Ratio (EAR) is a value that is used to measure eye openness by calculating the proportion of the vertical and horizontal distances of eye marks. When eyes are open the value of EAR is relatively high but when eyes start closing the vertical distance is lesser and therefore the value of EAR is very low. The system is able to monitor this ratio in consecutive video frames and identify indicators of drowsiness. In case the EAR reaches a preset value (usually approximately 0.25) during a certain number of consecutive frames, it is considered longer-term eye closure or micro-sleep and this triggers a drowsiness alarm (Soukupova & Cech, 2016).

1.3.2 MOUTH ASPECT RATIO (MAR)

The degree of mouth opening is measured by the Mouth Aspect Ratio (MAR), the measurement of the relative positions of important mouth landmarks. This measure is specifically effective in identifying yawning, which is one of the indicators of driver fatigue. The level of MAR when it is consistently high is an indication of long term mouth opening that is linked with yawning hence showing that the user may be drowsy. The application of MAR as an enhancement of the other measurements like the Eye Aspect Ratio (EAR) increases the system reliability and robustness of the system as it minimizes false alarms and offers a more detailed evaluation of driver fatigue (Gupta et al., 2018).

1.3.3 EMOTIONAL PROCESSING PIPELINE

The system also checks the emotional condition of the driver besides the physical indicators. In this project, we adopt a deep learning model with EfficientNet-B0 as it is selected due to its best tradeoff between accuracy and computational efficiency. In contrast to more bulky models, EfficientNet-B0 uses scaling of compounds to scale depth, width, and resolution uniformly to enable it to perform well in image classification tasks and yet be lightweight enough to make inferences in real-time on systems based on a CPU (Abdulkareem and Abdullah, 2025). This is especially applicable when it comes to continuous driver monitoring applications which demand low latency and resource efficiency.

With the help of this model, the system can categorize the facial expressions into the emotional conditions of stress, anger, or a neutral condition. Such classifications together with such physiological indicators of fatigue as eye closure and yawning can be used to better estimate the level of fatigue of the driver.

This assists in determining such situations as road rage or mental stress, which is also a contributor to unsafe driving.

1.3.4 THRESHOLD LOGIC AND EVENT DETECTION

All the measures obtained are compared with predetermined thresholds:

- EAR threshold → Eye closure detection
- MAR threshold → Yawning detection
- Emotion confidence → Stress detection

The system continuously tracks these values and triggers alerts when thresholds are exceeded for a certain duration.

1.3.5 ALERT SYSTEM AND HARDWARE INTEGRATION

Once it detects drowsiness, the system then sends alert immediately to alert the driver. Such notifications consist of audio notifications that are used to alert the driver and the screen displays notifications. These mechanisms combined make sure that the driver is alerted in time and in an effective manner.

1.3.6 LIVE SAFETY DASHBOARD

The system has a dashboard with real-time display of the state of the driver, which allows one to see a live video view, a drowsiness score between 0 and 100, event logs, and alert history. This

combined interface gives a chance to monitor everything continuously as well as deepen the analysis of the behaviour of the driver over time.

1.3.7 EMERGENCY SOS SYSTEM

In case a driver does not respond to the system in a long time, an emergency procedure is triggered, and an SOS notification is sent by Twilio. This alert is automatically sent to an emergency contact that has been pre-registered and will also have a clear distress message and the approximate GPS position of the driver so that they may identify where the person is. This feature will give an added protection to the safety and go a long way to improve the safety in situations where the driver is not able to reply and will greatly improve the situation in life threatening or critical situations.

1.3.8 TECHNOLOGIES USED

The system is built using:

- Python – Core programming
- OpenCV – Image processing
- Dlib – Facial landmark detection
- Flask – Web interface
- Scipy – Mathematical computations
- Arduino – Hardware integration
- Twilio API – Emergency communication

1.4 Challenges Faced

It is not simple to create a drowsiness detector that operates in real-time and is powered by an AI. A variety of technical and practical issues should be overcome.

1.4.1 LIGHTING CONDITIONS

Lighting variations is one of the most important issues in computer vision system. Face detection is also challenging in low-light conditions because they are not well visible and in situations where the lighting is too bright, it may hide some important faces. Also, shadows may deform features of the face, and it may be more difficult to identify important landmarks using the system accurately. Such differences in the lighting can dramatically decrease the overall accuracy and reliability of the detection process.

1.4.2 FACIAL OBSTRUCTIONS

Another significant weakness of the system is the facial obstructions. Driving drivers can be equipped with accessories like glasses, mask, or cap, which in a way cover the critical parts of the face. Specifically, the eyes and mouth, which are essential in revealing drowsiness, can be blurred. This means that the system will be incapable of recognizing facial landmarks and thus, will not be able to produce reliable, or consistent, detection results.

1.4.3 REAL-TIME PROCESSING CONSTRAINTS

The video frames must be processed in real time by the system and this presents the system with major computational challenges. Lag may be caused by high processing needs particularly when performing complex algorithms to detect and analyse. On the low-end hardware, the performance can slow down even more, leading to a slower response time or less accuracy. The optimal balance between processing speed and detection accuracy is one of the main problems in system design.

1.4.4 INDIVIDUAL DIFFERENCES

The other key problem is personal differences among users. Different people possess different structures of their face, such as the size of their eyes, the way they blink, and the amount of time they yawn. These differences ensure that it is hard to have universal limits when it comes to detecting drowsiness. Consequently, the system calibrated on one person may not automatically be as effective on another person, thus interfering with the consistency and reliability.

1.4.5 FALSE POSITIVES AND FALSE NEGATIVES

It is a complicated problem to make sure that the detection is accurate and not inconveniencing. Normal behaviours like blinking normally may be mistaken as drowsiness and false positives

obtained. In contrast, not all obviously tired drivers can have observable or noticeable symptoms, which makes false negatives. It is necessary to find a right balance between sensitivity and accuracy to keep the alerts unnecessary but at the same time remain safe.

1.4.6 EMOTIONAL DETECTION COMPLEXITY

The system is even more complicated with the detection of emotion. Facial expressions are very subtle and such emotions cannot be read accurately and there is also the likelihood of cultural differences in the way emotions are communicated. Moreover, some of the states like stress or fatigue may not necessarily leave any facial clues. The factors may cause wrong classification and decrease the efficiency of analysis based on emotions.

1.4.7 HARDWARE INTEGRATION ISSUES

When software is combined with hardware, e.g. Arduino, further problems arise. The responsiveness can be influenced by communication delays between hardware and the system, whereas sensor error can cause unreliable data. System stability can also be affected by power related problems. It is therefore important that there is a seamless and effective integration between the hardware and the software in a bid to assure quality operation.

1.4.8 PRIVACY CONCERNS

The privacy of the driver is a major factor since the system will use a camera to track the driver. Users might not be comfortable with the process of constant recording, and they might fear the possibilities of misuse of data that has been captured. To counter such concerns, one should make sure that all the data has to be processed locally and not stored or transmitted for no purpose therefore keeping all users and data safe.

1.4.9 NETWORK DEPENDENCY FOR SOS

The success use of the emergency SOS functionality is very much dependent on the network connectivity. In places where signal is weak or nonexistent, alerts might not transmit, or delay in transmitting. This drawback may greatly limit the reliability of the system especially in remote or rural areas where timely provision of help is of utmost importance.

1.4.10 SCALABILITY AND DEPLOYMENT

The implementation of the system on actual vehicles presents a number of practical issues. The system should be scalable to various camera systems and automobile conditions. It should also be able to deal with vibrations and movement and retain its level of detection. Also, it is necessary to guarantee long-term reliability and maintainability in different conditions, which will guarantee the successful implementation of large-scale implementation.

1.4.11 GPS INACCURACY

The GPS position does not present the actual location of the driver but it presents the ISP of the driver that can be quite dangerous in emergent circumstances.

1.5 PROJECT STATEMENT

Road safety has emerged as a significant issue over the past few years because of the growing number of accidents where driver fatigue and drowsiness led to accidents. As the transportation rapidly expands and people change their work patterns, the drivers are frequently expected to work long hours in physically and mentally hard conditions. Other causes like insufficient sleep, erratic timing, and stress greatly decrease alert and reaction time which influences the performance of driving.

Drowsiness is especially dangerous, as it builds up over a long period of time and is not always noticed by the driver. Compared to sudden distraction, fatigue is cumulative and drivers struggle to notice witnesses of the initial stages. Because of this, the drivers might think that they are fully attentive, whereas in reality, their concentration and awareness levels are decreasing. Conventional safety mechanisms largely depend on the driver consciousness or indirect signals like steering style and car movement.

Though these ways help to a certain extent, they are not completely dependable since they do not directly assess the physical and mental health of the driver. In most instances, these systems react when the performance of the driver has already been affected, thus limiting the effectiveness of the system in preventing accidents. To solve these weaknesses, we need a smart and proactive device able to track the driver on the road in real time and to identify the initial signs of fatigue. The proposed Driver Drowsiness Detection System meets this need by applying computer vision methods to monitor the behavior and face features of the driver.

Generally, the proposed system is intended to offer an efficient and convenient, non-invasive way of improving road safety. It assists in alleviating highway accidents that happen due to fatigue and helps to enhance safer driving conditions by integrating both real-time monitoring and various detection methods.

1.6 OBJECTIVES OF THE PROJECT

The main aim of the project is to create a smart and trustworthy driver monitoring system that will assist in detecting drowsiness in the driver-pilot in real time and inform him before it is too late to get into a hazardous situation. As the fatigue-related accidents become more frequent, it is necessary to develop a system that is able not only to identify drowsiness but also act upon it. The project aims at enhancing road safety through an unending evaluation of the facial features and behavioral patterns of the driver through the use of computer vision.

Key Objectives :

- To detect drowsiness with the EAR and MAR techniques.
- To track facial features as well as behavior in real-time.
- To add emotion detection to increase safety.
- To deliver immediate notifications using sound and devices.
- To introduce SOS emergency functionality.
- To provide low false detection and high accuracy.
- To develop an easy to use and affordable system.

One of the key aims of the system is to track the eye movements in terms of Eye Aspect Ratio (EAR). Human eye is one of the best indicators of alertness. As a driver gets fatigued, the rate at which he or she blink decreases and the eyes are left shut for a longer period. The system can be used to monitor such changes as well as to identify the signs of drowsiness at an early stage by computing the EAR. This assists in the identification of micro-sleep conditions which are very dangerous because they take place without the knowledge of the driver.

The other significant goal is to identify yawning based on the Mouth Aspect Ratio (MAR). One of the first signs of fatigue is often the yawning; particularly when driving long distances or at night. The system detects the extent of mouth opening and patterns that are linked with yawning. This, when used with eye detection, will offer a more credible way of determining drowsiness because the alternative will not be restricted to a single parameter.

Emotion detection is also a part of the monitoring system that will be introduced in the project. Bad emotions like stress, anger and frustration may have adverse influences on the driving behavior and predispose to accidents. Through facial expression, the system is able to tell whether the driver is emotionally stable or not. This further enhances safety and further makes the system more complete than simple drowsiness detection systems.

The other important goal is to provide real-time performance. The system will take the continuous video input and give instant feedback. This is very important since the system effectiveness can be diminished by any delay in detection. As such, the algorithms are designed to be efficient even on normal computing systems without having to acquire expensive hardware.

Another aspect of the project is the enhancement of accuracy and reduction of false detection. Existing conditions like lighting changes, head movement and background distraction may have an impact. The system is created to be able to cope with such challenges and deliver consistent results. One of the goals is to achieve a balance between sensitivity and accuracy.

Moreover, the system will combine hardware components to offer physical notifications. These are buzzer beeps and LED lights that instantly attract the driver. In conjunction with this, SOS feature is also adopted to send emergency messages and location information should the driver cease to respond.

1.7 SCOPE OF THE PROJECT

This project is a wide one that covers many fields of transportation and safety systems. As the demand of long-distance travel is growing and the number of vehicles on the road is increasing, there is a great necessity of intelligent systems that can help to increase the safety of drivers. This project offers a viable solution that can be implemented in the actual world with various classes of vehicles.

The system has the ability to be applied in both personal vehicles and commercial transportation like trucks and buses and ride sharing services. It particularly applies to the drivers who drive the vehicles over extended periods or at night shifts where the risks of fatigue are much greater. The system can also reduce the chances of accidents due to drowsiness by ensuring that there is constant monitoring.

Functionally, the system conducts real time monitoring with a camera. It recognizes the face of the driver and determines the facial features to interpret the movement of the eyes and mouth. The system can identify the alertness or drowsiness of the driver by computing EAR and MAR values. These characteristics render the system to be an active safety mechanism.

Emotion detection also increases the scope of the project. The system can also identify stress or aggressive behavior by analyzing the facial expressions, which is also a significant factor in road safety. This renders the system more holistic and efficient.

Technically, the project incorporates several technologies, including computer vision, deep learning, and embedded systems. Image processing and facial landmark detection are performed with the tools such as OpenCV and Dlib, whereas emotion recognition is performed with deep learning models. A Python-based backend is used to control the system, and alerts are done with hardware that uses Arduino.

The project is planned to be feasible and economical. It does not need costly sensors or advanced machines since it is based primarily on a camera and simple computing equipment. This enhances its practicality and renders it applicable to real life application.

Scope Highlights:

1. Suited in cars, trucks, buses, and taxis.
2. Real-time work with camera surveillance.
3. Integrates both physical and emotional sensing.
4. Utilizes low cost and available technology.
5. Does not require intrusive sensors.

Future Scope:

The project holds a lot of potential concerning future improvements and expansion. As technologies evolve, the system can also be further improved to increase performance and usability.

1. Night vision through integration with infrared cameras.
2. Cloud storage and analysis of data.
3. Creation of mobile apps to monitor remotely.
4. Connection with IoT and smart cars.
5. Predictive fatigue using advanced AI models.

In general, this project is not confined to a prototype but has real-world applications and future technological advances.

1.8 LIMITATIONS OF THE PROJECT

Although the proposed driver drowsiness detection system is very effective, it has many limitations that must be put into consideration in order to be put into practical use. Among the main weaknesses is the fact that the system relies on the lighting conditions. Because the system provides a camera to scan and scan facial features, bad lighting conditions like driving at night or extreme brightness may decrease the image quality and compromise the quality of scan.

The other significant limitation is that of clear and consistent view of the face of the driver. Under real-world circumstances, there are factors that may disrupt the appropriate detection, they include:

- Wearing sunglasses or masks
- Repeat movement of the head or turning of the face.
- Poor positioning of the camera.
- Half blockage of features of face.

Such factors may lower the precision of the identification of eye and mouth movements which are crucial in determining drowsiness.

Emotion detection is also subject to some issues with its accuracy. The expression of the human face differs greatly among people and the expression itself can be taken to indicate a different mood as well. Consequently, the system can sometimes fail to interpret emotions and this can influence the credibility of the entire analysis.

The other constraint is associated with computer needs. The system is based on the constant video processing, which may be intensive, in particular, when many features are considered at the same time. This may lead to:

- Heavy processing load on the system.
- Delays in real-time detection.
- Less performance with low-end devices.

Moreover, the system relies primarily on visual cues and lacks such physiological parameters as heart rate or brain signals that would enhance accuracy. Nevertheless, the process of implementing such features would complicate and raise the cost of the system.

On the whole, even despite some limitations of the system, it should also be considered a great base of enhancing the safety of drivers. These limitations can be solved in future development with further development and incorporation of other technologies.

Chapter 2

Literature Review

2.1 INTRODUCTION

The practice of driver drowsiness detection has developed, not only as an expansion of intrusive physiological sensor technologies (EEG, EOG, ECG), but also as non-contact vision-based technologies that derive fatigue indicators directly out of facial video. The systems measure the period of eye closure, frequency of blinks as well as yawning and head orientation using computer vision methods, both traditional geometric measures and deep learning classifiers. The survey below investigates the most important contributions which would be a theoretical basis of this project.

2.2 SURVEY ON DRIVER DROWSINESS DETECTION

2.2.1 EAR-BASED EYE CLOSURE DETECTION

Soukupova and Cech (2016) proposed the Eye Aspect Ratio (EAR) to real-time detect eye blink and eye closure with the help of the 68-point facial landmark predictor of dlib. The EAR is calculated with six landmark points surrounding each eye and decreases drastically when the eye is closed. The method uses reliable data to differentiate between blinks and long-term closures in-

the-field despite the lack of any drowsiness-specific training data. Its major weakness is its sensitivity to the camera angle and ambient light.

$$\text{EAR} = (\|P1 - P5\| + \|P2 - P4\|) / (2 \times \|P0 - P3\|)$$

King (2009) described the dlib machine learning toolkit, the ensemble of regression trees of which is the 68-point landmark predictor that is used in EAR computation. The predictor has almost real-time performance and is generalizable to a wide variety of facial appearances and moderate pose variations.

2.2.2 HAAR CASCADE CLASSIFIERS

Viola and Jones (2001) proposed the Haar Cascade framework for rapid multi-scale face detection based on AdaBoost-trained feature cascades. Its speed and easy accessibility through OpenCV render it the default first-stage face detector in drowsiness pipelines. The weaknesses are low precision with extreme head positions and low brightness.

2.2.3 THE PERCLOS METRIC

Wierwille and Ellsworth (1994) validated the Percentage of Eye Closure (PERCLOS) measure, which is the percentage of time when the eye is over 80 percent closed within a sliding window, as an effective measurement of drowsiness. Dinges et al. (1998) once again confirmed that there were good correlations between PERCLOS and psychomotor vigilance degradation, thus making it an industry-standard fatigue monitoring system.

2.2.4 DEEP LEARNING APPROACHES

Weng, Lai, and Lai (2017) used a Hierarchical Temporal Deep Belief Network on the image sequences of eye regions and were able to perform better than EAR-based methods in challenging lighting conditions but at the cost of high computation. Redmon and Farhadi (2018) showed that the throughput of the single-pass detection of YOLOv3 is higher than the cascade-based ones, but fine-grained facial subregions are still a weakness.

2.2.5 HEAD POSE ESTIMATION

Hossain, Islam, and Ali (2020) integrated the EAR-based eye-tracking with the OpenCV solvePnP head pose estimation (returns roll, pitch, and yaw angles) to enhance the robustness of drowsiness classification compared to single-modality systems. Constant downward pitch combined with low EAR was found to produce significant false negatives, but night-time performance was reported to be impaired.

2.2.6 HYBRID HARDWARE-SOFTWARE SYSTEMS

Saini, Kumar, and Mittal (2021) demonstrated that an Arduino-based system (with 89 percent detection accuracy and serial alert latency of less than 150 ms) confirms that commodity microcontrollers provide a feasible hardware alerting layer. According to Jabbar, Islam, and Alam (2018), the prevailing factors of accuracy in embedded deployments were camera quality and lighting.

2.2.7 EMOTION RECOGNITION IN DRIVER MONITORING

The universal taxonomy of six expressions formalised in FACS, developed by Ekman and Friesen (1978), forms the basis of the eight-emotion model deployed by EmotiEFfLib in this project. It was demonstrated that FACS-based classifiers depict trustworthy real-time emotion recognition and that negative emotions like anger, disgust, and fear are associated with compromised driving behaviour — this justifies their application as complementary drowsiness indicators (Zeng et al., 2009).

2.2.8 COMPARATIVE SUMMARY OF LITERATURE

Table 2.1: Comparison of Reviewed Works on Driver Drowsiness Detection

Author(s) & Year	Method / Tool	Key Finding	Limitation
Soukupova & Cech (2016)	EAR with dlib 68-pt landmarks	Reliable real-time blink detection without training data	Sensitive to camera angle and lighting
Viola & Jones (2001)	Haar Cascade (AdaBoost)	Fast multi-scale face detection in real time	Poor at extreme head orientations and low light
Wierwille & Ellsworth (1994)	PERCLOS metric	Validated drowsiness indicator in simulator studies	Susceptible to blink rate variability and occlusion
Dinges et al. (1998)	PERCLOS & psychomotor vigilance	Strong correlation with reaction time degradation	Requires continuous accurate eye tracking
King (2009)	dlib shape predictor (regression trees)	Real-time 68-pt landmark accuracy across diverse faces	Accuracy drops under heavy occlusion
Weng et al. (2017)	Hierarchical Temporal Deep Belief Network	Outperforms EAR under challenging lighting	High computational demand, unsuitable for low-spec hardware
Hossain et al. (2020)	EAR + solvePnP head pose	Improved robustness over EAR-only systems	Performance drop in night-time conditions
Saini et al. (2021)	OpenCV EAR + Arduino buzzer/LED	Serial alert latency < 150 ms; 89% accuracy	Tested in simulated conditions only
Zeng et al. (2009)	Facial Action Unit classification	Real-time emotion recognition for driver monitoring	Performance varies with pose and illumination

Chapter 3

Methodology

This chapter explains the end-to-end approach of the Driver Drowsiness Detection System, which is subdivided into a software processing pipeline and a hardware integration layer. A complete real-time detection and alerting system is created by the two subsystems communicating via a serial interface.

3.1 SOFTWARE PIPELINE

3.1.1 FACE DETECTION

Each video frame is converted to greyscale and scanned using OpenCV's Haar Cascade frontal face classifier (haarcascade_frontalface_default.xml) with a scale factor of 1.1, minimum neighbours of 5, and minimum size of 30x30 pixels. An independent alert flag is raised if no face is detected over 30 consecutive frames (approximately 1 second at 30 fps).

3.1.2 FACIAL LANDMARK EXTRACTION

Dlib's shape_predictor_68_face_landmarks.dat model extracts 68 landmark coordinates from the detected face area. Eye landmarks (indices 36–47) and mouth landmarks (48–67) are stored as NumPy arrays for downstream computation.

3.1.3 EYE ASPECT RATIO COMPUTATION

EAR is calculated per eye based on the Soukupova-Cech formula and averaged across both eyes. Transient noise is smoothed with a rolling window of five frames. When the average EAR falls below 0.25 for three consecutive frames, the EAR drowsiness flag is raised.

$$\text{EAR} = (\|P1 - P5\| + \|P2 - P4\|) / (2 \times \|P0 - P3\|)$$

3.1.4 MOUTH ASPECT RATIO AND YAWN DETECTION

The Mouth Aspect Ratio (MAR) is computed as the ratio of mean vertical mouth opening to the horizontal width of the mouth. A MAR value greater than 0.6 sustained over more than ten successive frames is classified as a yawn event and increments the yawn counter.

3.1.5 HEAD POSE ESTIMATION

The solvePnP function of OpenCV correlates the points of a 3D model of the face with their 2D landmarks, providing rotation and translation vectors. Through the Rodrigues formula, Euler angles (pitch, yaw, roll) are obtained. Any downward pitch that is more than 20 degrees over 15 frames in succession raises the head-pose drowsiness flag.

3.1.6 EMOTION RECOGNITION

EmotiEFfLib recognizes the facial expression of the driver in real time and classifies it in eight emotions. Emotions that are considered to be high risk (Sad, Anger, Disgust, Fear, Contempt) advance a risk counter, whereas the less risky emotions (Happy, Neutral, Surprise) do not. The risk counter has the side effect of adding a secondary weight to the alert scoring formula. Table 3.1 shows the full mapping.

Table 3.1: Emotion-to-Risk Level Mapping (EmotiEFfLib)

Emotion	Risk Level	Rationale
Happy	LOW RISK	Positive affect; associated with alert, engaged state
Neutral	LOW RISK	Baseline driving state; no impairment indicator
Surprise	LOW RISK	Heightened attention; not indicative of fatigue
Sad	HIGH RISK	Linked to reduced arousal and cognitive engagement
Anger	HIGH RISK	Cognitive distraction; impairs hazard anticipation
Disgust	HIGH RISK	Negative affect associated with distraction
Fear	HIGH RISK	Panic response; degrades rational decision-making

Contempt	HIGH RISK	Disengaged affect; correlated with inattention
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3.1.7 ALERT FLAG GENERATION

A weighted scoring formula aggregates the four module outputs: the EAR flag (weight 0.50), the yawn counter (0.25), the head pose flag (0.15), and the emotion risk counter (0.10). An alert is sent when the score exceeds 0.5. A hysteresis mechanism requires the score to fall below 0.35 for at least 20 frames before the alert is cleared.

3.2 HARDWARE INTEGRATION

The Arduino Uno is connected to the host PC via USB serial at 9600 baud. On receiving character 'A' (Alert), the Arduino displays 'DROWSY! PULL OVER' on the 16x2 LCD and pulses the buzzer (500 ms on / 200 ms off) for three seconds. On receiving 'N' (Normal), it resets to standby. All hardware components are outlined in Table 3.2.

Table 3.2: Hardware Components and Specifications

Component	Specification	Role in System
Arduino Uno	ATmega328P, 16 MHz	Microcontroller; receives serial alert commands from Python
USB Camera	640x480, 30 fps	Captures real-time facial video stream
16x2 LCD Display	HD44780 controller	Displays driver status and alert messages
Active Buzzer (5V)	85 dB, 2.5 kHz	Produces audible alarm on drowsiness detection
USB Serial Cable	USB-A to USB-B	Serial communication between host PC and Arduino

3.3 SYSTEM FLOWCHART

On start-up, the camera and Arduino serial link are initialised. Each frame is greyscaled and face-detected. If no face is found, the no-face counter increments toward its alert threshold. If a face is found, EAR, MAR, head pose, and emotion outputs are computed in parallel and fed into the weighted scorer. A score above 0.5 transmits 'A' to the Arduino; otherwise 'N' is sent. The hysteresis mechanism prevents rapid alert toggling. The loop repeats until the application terminates.

Chapter 4

IMPLEMENTATION

The driver drowsiness detection system is built using Python and works in real time. A webcam is placed in front of the driver to continuously capture facial movements while driving. Every frame from the webcam is processed immediately so that the system can quickly identify signs of tiredness.

The first step is face detection. Using OpenCV and Dlib, the system detects the driver's face and marks 68 facial landmark points. These points are mainly around the eyes, mouth, nose, and jawline. Among these, the eyes and mouth are the most important because they help identify drowsiness.

OpenCV is used to capture the driver's live video feed and process each frame in real time. Dlib helps the system detect the driver's face and locate facial landmark points. The system mainly focuses on the eyes and mouth because they are the most important indicators of drowsiness. Eye Aspect Ratio (EAR) is used to measure whether the driver's eyes are open or closed. If the eyes stay closed for too long, the system treats it as a sign of fatigue or sleepiness. Mouth Aspect Ratio (MAR) is used to check whether the driver is yawning. If the mouth remains open for a longer period, the system identifies it as a yawn. EAR and MAR values are monitored continuously so that the system can detect drowsiness more accurately and quickly.

Technologies Used

- Python
- OpenCV
- Dlib
- Flask
- Scipy
- Arduino
- Twilio API

Overall, the implementation combines computer vision, AI, and hardware integration to create a reliable driver safety.

The implementation phase involves translating the conceptual architecture into a functional Python-based application. This section details the algorithmic flow, data processing techniques, and the integration of deep learning models for real-time driver monitoring.

4.1 SOFTWARE ENVIRONMENT AND LIBRARIES

The system was developed using **Python 3.x** due to its extensive support for computer vision and machine learning.

- **OpenCV:** Used for real-time video stream acquisition and image preprocessing (grayscale conversion and resizing).
- **Dlib:** Implemented for its pre-trained **68-point facial landmark predictor**, which provides the localized coordinates needed for ratio calculations.
- **NumPy & SciPy:** Utilized for high-speed vector calculations, specifically for calculating Euclidean distances between landmark points.
- **TensorFlow/Keras:** Used to load the pre-trained **EfficientNet-B0** model for emotion classification.

4.2 ALGORITHMIC WORKFLOW

The implementation follows a sequential pipeline executed for every frame captured by the webcam:

1. **Frame Acquisition:** Captures a frame and converts it to grayscale to reduce computational overhead for landmark detection.
2. **Face Localization:** Dlib's HOG-based face detector identifies the bounding box of the driver's face.
3. **Landmark Mapping:** The 68-point predictor maps specific coordinates. For the eyes, indices [36–41] (left) and [42–47] (right) are extracted. For the mouth, indices [48–67] are targeted.

4.3 CODING THE MATHEMATICAL MODELS

4.3.1 EYE ASPECT RATIO (EAR) LOGIC

The EAR is implemented by calculating the distance between vertical eye landmarks and dividing it by the distance between horizontal landmarks.

If the EAR remains below the threshold of **0.25** for more than **48 consecutive frames** (approx. 2 seconds), the system flags a "Drowsiness Event."

4.3.2 MOUTH ASPECT RATIO (MAR) LOGIC

To detect yawning, the MAR calculates the opening of the lips.

Unlike the EAR, the MAR threshold is set higher (approx. **0.6**). If this threshold is crossed, the system increments a "Yawn Counter" on the dashboard.

4.4 EMOTION CLASSIFICATION PIPELINE

We implemented **EfficientNet-B0** for its "Compound Scaling" property, which allows it to run efficiently on a standard CPU without requiring a dedicated GPU.

- **Preprocessing:** The detected face is cropped, resized to **224x224 pixels**, and normalized.
- **Inference:** The model predicts probabilities for seven emotions. We specifically monitor "Stress" and "Anger" as indicators of unsafe driving behavior.
- **Integration:** The emotion result is overlaid on the video feed and logged into the daily safety report.

4.5 HARDWARE AND EXTERNAL INTEGRATION

4.5.1 ARDUINO AND SERIAL COMMUNICATION

The Python script communicates with the **Arduino Uno** via the PySerial library.

- When a drowsiness event is confirmed, Python sends a signal (e.g., '1') over the COM port.
- The Arduino receives this signal and triggers a **5V Piezo Buzzer** and a Red LED.

4.5.2 EMERGENCY SOS (TWILIO API)

The SOS module is implemented as a "fail-safe" function. If the driver remains in a "Drowsy" or "Closed Eyes" state for more than **10 seconds**, the system calls the **Twilio REST API**.

- **Data Sent:** A pre-defined SMS message containing the driver's last known status.
- **Geolocation:** Using the geocoder library, the system fetches the current IP-based GPS coordinates and appends a Google Maps link to the alert message.

4.6 USER INTERFACE (LIVE DASHBOARD)

The front-end is implemented using **Flask**, creating a local web server.

- **Video Streaming:** Uses multipart/x-mixed-replace responses to stream the processed OpenCV frames to a web browser.

- **Telemetry:** A side panel displays real-time EAR/MAR values allowing for visual monitoring of the driver's fatigue trends.

Chapter 5

RESULTS AND DISCUSSION

5.1 SYSTEM PERFORMANCE AND KEY FINDINGS

The experimental evaluation of the Driver's Drowsiness Detection System confirms that a multi-modal approach—combining physiological and psychological indicators—is significantly more reliable than single-parameter systems. The following results were observed during the testing phase:

- **Precision in Fatigue Detection:** The system successfully utilized **EAR (Eye Aspect Ratio)** to distinguish between natural blinking and micro-sleep. By requiring the EAR to remain below the 0.25 threshold for a specific frame duration, we effectively minimized false positives from rapid blinking.
- **Behavioural Correlation:** The integration of **MAR (Mouth Aspect Ratio)** proved essential. In several test cases, drivers exhibited yawning (high MAR) several minutes before their EAR values indicated a micro-sleep, validating yawning as a critical early-warning sign.
- **Real-Time Responsiveness:** The system maintained a consistent processing speed, ensuring that audio-visual alerts and the **Arduino-based buzzer** were triggered within milliseconds of a detected event.
- **Safety Redundancy:** The **Emergency SOS module** functioned as a vital fail-safe. During unresponsive simulations, the Twilio API successfully dispatched distress messages with embedded GPS coordinates, proving the system's utility in high-risk scenarios.

5.2 DISCUSSION OF CHALLENGES AND LIMITATIONS

While the system proved highly effective in controlled environments, the testing phase highlighted several real-world variables that impact performance:

5.2.1 Environmental and Hardware Constraints

The primary challenge encountered was **lighting variability**. Standard webcams struggled to isolate facial landmarks in low-light or night-driving conditions, which are precisely when the system is most needed. Furthermore, the **Emergency SOS feature** is currently dependent on active internet connectivity; in remote areas with poor cellular reception, the delivery of SMS alerts may be delayed or fail.

5.2.2 Physical and Behavioural Occlusions

The accuracy of the Dlib 68-point predictor was occasionally compromised by external factors:

- **Accessories:** Items such as thick-rimmed glasses, medical masks, or low-hanging caps can obscure the eyes and mouth, leading to missed detections.
- **Head Pose:** Sudden or extreme head movements (looking far left or right) sometimes caused the system to lose the facial "lock," momentarily interrupting the monitoring stream.

5.2.3 Algorithmic Nuances

A notable observation was the "Silent Fatigue" phenomenon, where a driver may feel internally exhausted without exhibiting overt signs like yawning or prolonged eye closure. Additionally, while the system is calibrated for general use, individual variations in eye shape and blinking frequency mean that a "one-size-fits-all" threshold can occasionally trigger unnecessary alerts for some users.

5.3 COMPARATIVE ADVANTAGE

Despite these limitations, the system represents a significant step forward in low-cost road safety. By analysing **Eye, Mouth, and Emotion** data simultaneously, the system provides a holistic view of the driver's state. This layered approach ensures that if one indicator is masked (e.g., if the driver is wearing sunglasses but yawning), the system can still identify the danger.

Chapter 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The development of this **Driver's Drowsiness Detection System** serves as a practical demonstration of how modern advancements in Computer Vision and Artificial Intelligence can be directly applied to save lives. Fatigue remains one of the leading causes of road accidents globally, and this project successfully addresses the limitations of traditional, indirect safety measures.

By implementing a **multi-parameter detection architecture**, the system moves beyond simple eye-tracking. The integration of **Eye Aspect Ratio (EAR)**, **Mouth Aspect Ratio (MAR)**, and **Emotion Analytics** creates a robust safety net that identifies the earliest physiological and psychological signs of exhaustion. Our research and testing have shown that:

- **Precision matters:** Combining multiple indicators significantly reduces false alarms compared to systems relying on a single metric.
- **Immediate Feedback is Crucial:** The synergy between Python-based software and **Arduino-based hardware** ensures that alerts are not just seen, but felt, providing the necessary stimulus to snap a driver out of a micro-sleep.
- **Safety is Holistic:** The inclusion of the **Emergency SOS feature** transforms the system from a mere monitoring tool into an active rescue device, providing critical location data when the driver is incapacitated.

Ultimately, this project proves that a low-cost, non-intrusive monitoring solution is not only feasible but essential for the future of intelligent transportation.

6.2 FUTURE WORK

While the current system provides a strong foundation for driver safety, the following enhancements are proposed to transition the project from a functional prototype to an industry-standard safety module:

6.2.1 Environmental and Sensory Upgrades

- **Infrared (IR) Integration:** To overcome the challenges of low-light and night driving, future iterations should incorporate IR cameras and emitters to ensure landmark detection remains accurate in total darkness.
- **Advanced Geolocation:** Future work will focus on refining GPS tracking algorithms to provide higher-precision coordinates, potentially integrating with **Cellular V2X (Vehicle-to-Everything)** technology for faster emergency response.

6.2.2 Active Vehicle Control

- **Direct Hardware Intervention:** Beyond simple alerts, the system could be interfaced directly with a vehicle's **Electronic Control Unit (ECU)**. In critical scenarios where a driver is unresponsive, the system could automatically engage hazard lights, reduce engine speed, or initiate a controlled emergency stop.

6.2.3 Expanding the behavioural Model

- **Multi-Modal AI:** Future models could be trained to analyze additional indicators such as **head-nodding frequency**, **steering wheel grip pressure**, and **voice pitch analysis** to detect cognitive fatigue that may not be visible on the face.
- **Occlusion Robustness:** We aim to implement more sophisticated deep learning models capable of "filling in" facial data when the driver is wearing masks, sunglasses, or caps, ensuring 100% uptime regardless of the driver's attire.

6.2.4 Cloud and Fleet Management

- **IoT Dashboarding:** Transitioning the local dashboard to a cloud-based platform would allow logistics and fleet managers to monitor driver wellness across an entire fleet in real-time, enabling proactive scheduling and rest-break management.

CONCLUSION

The *AI-Driven Driver's Drowsiness Detection System* successfully demonstrates how modern technologies such as computer vision, deep learning, and embedded systems can be integrated to address a critical real-world problem—road accidents caused by driver fatigue. By combining multiple detection parameters including Eye Aspect Ratio (EAR), Mouth Aspect Ratio (MAR), head pose estimation, and emotion recognition, the system provides a comprehensive and reliable method for identifying early signs of drowsiness.

The implementation of a real-time monitoring pipeline ensures timely detection and immediate alert generation, significantly reducing the chances of delayed response. The integration of hardware components such as Arduino-based buzzers and visual indicators enhances the effectiveness of alerts by engaging multiple sensory channels. Additionally, the inclusion of an Emergency SOS system adds a crucial layer of safety, enabling assistance in critical situations where the driver becomes unresponsive.

Experimental results indicate that the multi-modal approach improves detection accuracy and minimizes false positives compared to traditional single-parameter systems. While certain limitations such as lighting conditions, facial obstructions, and dependency on network connectivity were observed, the system provides a strong foundation for further improvements.

In conclusion, this project highlights the potential of AI-based non-intrusive monitoring systems in enhancing road safety. With future enhancements such as infrared imaging, cloud integration,

and advanced AI models, the system can evolve into a scalable and industry-ready solution for intelligent transportation systems.

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